CampusPartner: An Assistive Technology for Mobility Impaired Pedestrians

Department of Computer Science

Problem

When creating a pedestrian route in widely used navigation apps, there is not sufficient customization provided to users in terms of obstacles they want to avoid.

- These navigation apps are not adequate for people with mobility impairments (MIs).
- Accessibility issues are especially present on college campuses in the U.S.
- People with MIs may need to plan ahead, researching building entrances, directions, and other accommodations.

Solution

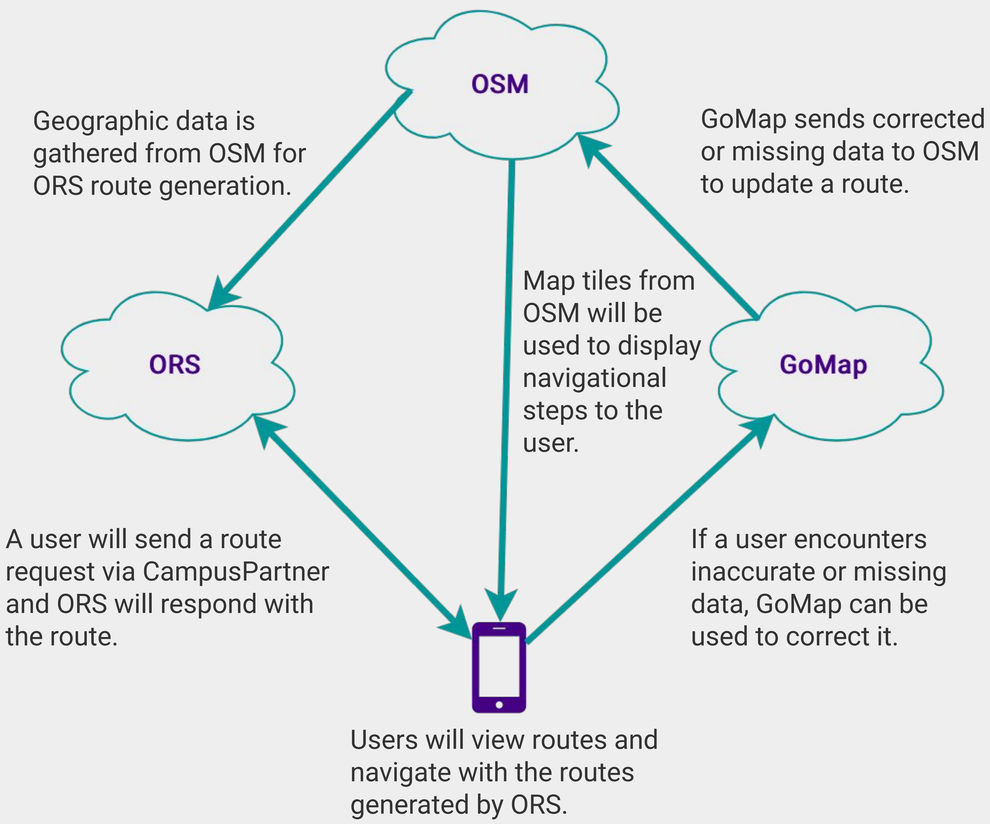
CampusPartner will assist people with MIs in viewing and planning walking routes in advance.

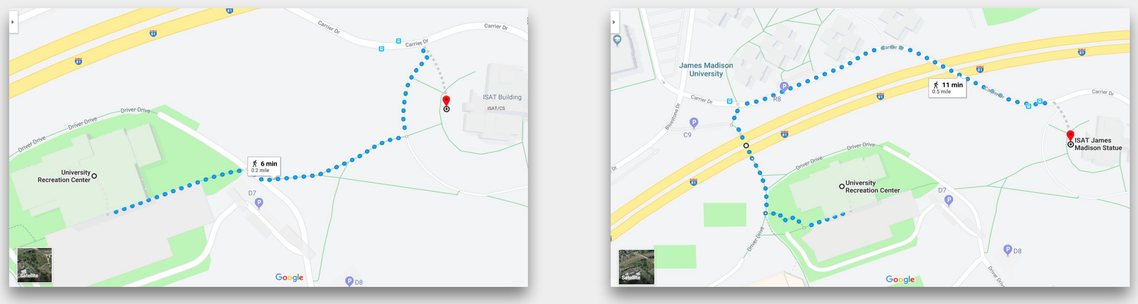
Supports users in:

- Creating a profile to inform route creation.
- Viewing routes in detail with turn-by-turn directions.
- Saving routes for quick access later.
- Updating missing or inaccurate information on the map.

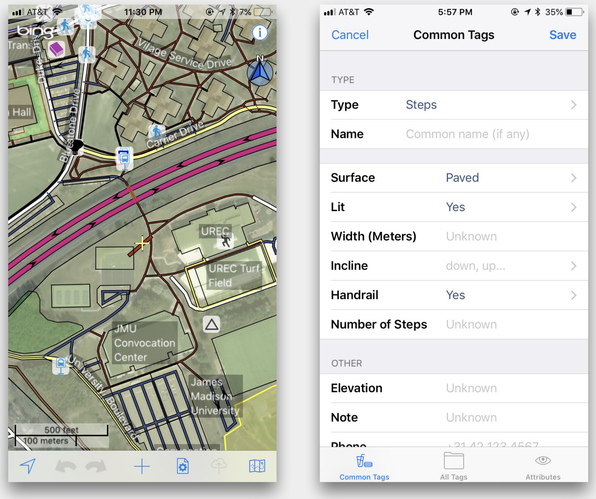
CampusPartner uses OpenStreetMaps (OSM) and OpenRouteService (ORS) in route generation and visualization.

CampusPartner





The fastest walking route versus the accessible route drawn in Google Maps.



The map and editing interfaces of GoMap when adding missing stairs to OSM.

Collaborative Design

CampusPartner is designed to:

- Ally the user with the assistive technology and with past and future users to find and verify the accuracy of the routes.
- Leave space for the users' sense of agency by positioning users to contribute to the data source they are using within the app.

Future Work

User studies -

Once an initial prototype is completed, the app will be tested in collaboration with users with MIs.

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http://tiny.cc/CampusPartner

References

Please find our references in our extended abstract.